An Analysis of Adversarial machine learning

Adversarial machine learning

Adversarial machine learning -- Part 1

argmax k = 1 , . . . , K f k (x ^) ≠ y , | | x ^ − x | | p ≤ ■ and x ^ ∈ [0 , 1] d {\displaystyle {\text{argmax}}_{k=1,...,K}f_{k}({\hat {x}})\neq y,||{\hat {x}}-x||_{p}\leq \epsilon {\text{ and }}{\hat {x}}\in [0,1]^{d}}

Adversarial machine learning -- Part 2

Specific attack types There are a large variety of different adversarial attacks that can be used against machine learning systems. Many of these work on both deep learning systems as well as traditional machine learning models such as SVMs and linear regression. A high level sample of these attack types include:

Adversarial machine learning -- Part 3

Evasion Evasion attacks consist of exploiting the imperfection of a trained model. For instance, spammers and hackers often attempt to evade detection by obfuscating the content of spam emails and malware. Samples are modified to evade detection; that is, to be classified as legitimate. This does not involve influence over the training data. A clear example of evasion is image-based spam in which the spam content is embedded within an attached image to evade textual analysis by anti-spam filters. Another example of evasion is given by spoofing attacks against biometric verification systems. Evasion attacks can be generally split into two different categories: black box attacks and white box attacks.

Adversarial machine learning -- Part 4

Untargeted: min x ′ d (x ′ , x) subject to C (x ′) ≠ C (x) {\displaystyle {\textbf {Untargeted:}}\min _{x^{\prime }}d(x^{\prime },x){\text{ subject to }}C(x^{\prime })\neq C(x)}

Adversarial machine learning -- Part 5

Black box attacks Black box attacks in adversarial machine learning assume that the adversary can only get outputs for provided inputs and has no knowledge of the model structure or parameters. In this case, the adversarial example is generated either using a model created from scratch, or without any model at all (excluding the ability to query the original model). In either case, the objective of these attacks is to create adversarial examples that are able to transfer to the black box model in question.

Adversarial machine learning -- Part 6

S (x ′) > 0 ■ { a r g m a x c F (x ′) ≠ C (x) , (Untargeted) a r g m a x c F (x ′) = c ∗ , (Targeted) {\displaystyle S(x^{\prime })>0\iff {\begin{cases}argmax_{c}F(x^{\prime })\neq C(x),&{\text{(Untargeted)}}\\argmax_{c}F(x^{\prime })=c^{*},&{\text{(Targeted)}}\end{cases}}}

Adversarial machine learning -- Part 7

Performance evaluation The choice of the evaluation metrics can impact the validity of the results. Having an inappropriate baseline involves failing to compare a new model against simpler, well-established baselines. Inappropriate performance measures means using metrics that do not align with the practical goals of the system. Reporting only "accuracy" is often described as insufficient for an intrusion detection system, where false-positive rates are considered critically important. Base rate fallacy is a failure to correctly interpret performance in the context of large class imbalances.

Adversarial machine learning -- Part 8

min (| | δ | | p + c ⋅ f (x + δ)) , x + δ ∈ [0 , 1] n {\displaystyle \min(||\delta ||_{p}+c\cdot f(x+\delta)),x+\delta \in [0,1]^{n}}

Adversarial machine learning -- Part 9

Fast gradient sign method One of the first proposed attacks for generating adversarial examples was proposed by Google researchers Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. The attack was called fast gradient sign method (FGSM), and it consists of adding a linear amount of in-perceivable noise to the image and causing a model to incorrectly classify it. This noise is calculated by multiplying the sign of the gradient with respect to the image we want to perturb by a small constant epsilon. As epsilon increases, the model is more likely to be fooled, but the perturbations become easier to identify as well. Shown below is the equation to generate an adversarial example where x {\textstyle x} is the original image, ■ {\textstyle \epsilon } is a very small number, Δ x {\textstyle \Delta _{x}} is the gradient function, J {\textstyle J} is the loss function, θ {\textstyle \theta } is the model weights, and y {\textstyle y} is the true label.

Adversarial machine learning -- Part 10

adversarial training of a linear regression model with input perturbations restricted by the infinity-norm closely resembles Lasso regression, and that adversarial training of a linear regression model with input perturbations restricted by the 2-norm closely resembles Ridge regression.

Adversarial machine learning -- Part 11

Adversarial natural language processing Adversarial attacks on speech recognition have been introduced for speech-to-text applications, in particular for Mozilla's implementation of DeepSpeech and Google's Speech Commands model. Various pre-processing techniques have been proposed to defend against such attacks , but these methods may not be resilient to attackers aware of those defenses.

Adversarial machine learning -- Part 12

Adversarial deep reinforcement learning Adversarial deep reinforcement learning is an active area of research in reinforcement learning focusing on vulnerabilities of learned policies. In this research area, some studies initially showed that reinforcement learning policies are susceptible to imperceptible adversarial manipulations. While some methods have been proposed to overcome these susceptibilities, in the most recent studies it has been shown that these proposed solutions are far from providing an accurate representation of current vulnerabilities of deep reinforcement learning policies.

Adversarial machine learning -- Part 13

One important property of this equation is that the gradient is calculated with respect to the input image since the goal is to generate an image that maximizes the loss for the original image of true label y {\textstyle y} . In traditional gradient descent (for model training), the gradient is used to update the weights of the model since the goal is to minimize the loss for the model on a ground truth dataset. The Fast Gradient Sign Method was proposed as a fast way to generate adversarial examples to evade the model, based on the hypothesis that neural networks cannot resist even linear amounts of perturbation to the input. FGSM has shown to be effective in adversarial attacks for image classification and skeletal action recognition.

Adversarial machine learning -- Part 14

Deployment and operation Deployment introduces challenges related to the performance and security in live environments. Lab-only evaluation is the practice of evaluating a system only in a controlled, static laboratory setting, which does not account for real-world challenges like concept drift and performance overhead. Inappropriate threat model refers to failing to consider the ML system itself as an attack surface.